INCH-POUND

MIL-C-3098/133C
24 September 1992
SUPERSEDING
MIL-C-3098/133B
9 November 1983

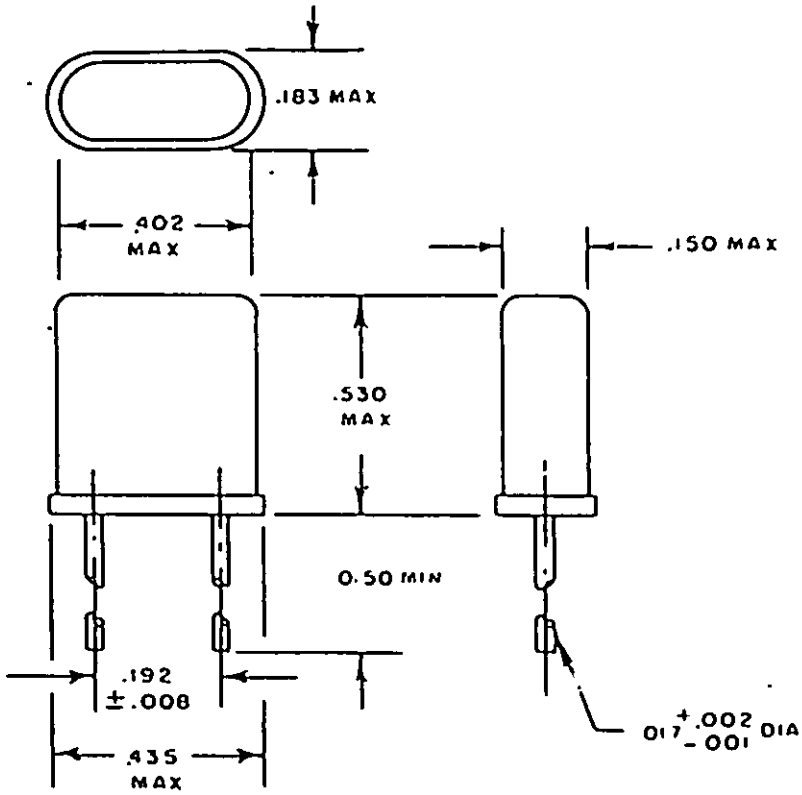
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR152/U

This specification is approved for use by all Department and Agencies of the Department of Defense.

The requirement for the acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-3098.

Pertinent characteristics: 16 to 61 MHz; third mechanical overtone; noncontrolled; series resonance.



Inches	mm
.001	0.03
.002	0.05
.008	0.20
.017	0.43
.150	3.81
.183	4.65
.192	4.88
.402	10.21
.435	11.05
.50	12.7
.530	13.46

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking shall be in accordance with MIL-C-3098.

FIGURE 1. Crystal unit - CR152/U.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:59Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Holder: HC-18/U or HC-43/U (MIL-H-10056/09).

Dimensions, marking, and configuration: See figure 1

Frequency range: 16.0 to 61 MHz, inclusive

Frequency tolerance:

Primary operating temperature range: ±20 ppm.

Secondary operating temperature range: ±30 ppm.

Equivalent resistance: 40 ohms maximum.

Mode of oscillator: Third mechanical overtone.

Operating temperature ranges:

Primary: -40°C to -90°C, inclusive.

Secondary: -55°C to -40°C and +90°C to +105°C, inclusive.

Drive level: 2.0 mW maximum.

Capacitance, shunt: 3.5 pF maximum.

Unwanted modes: Method II, minimum unwanted-to-main mode resistance ratio of 3:1.

Shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Vibration: Method 204 of MIL-STD-202, test condition A.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Aging:

Frequency change permitted: ±5 ppm.

NOTE: These crystal units for use in the AN/ARC-101 equipment. They were previously fabricated using AVCO Corporation Specification 809051-1 through -10.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:59Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI, SM
Navy - SH
Air Force - 19
DLA - ES

User activities:
Navy - AS, CG, MC, OS
Air Force - 17, 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5955-0672-06)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:59Z
Check the source to verify that this is the current version before use.